CARIS LIFE SCIENCES

Patient

Name: CARRIGAN, GARY L
Date of Birth: 06/19/1950
Sex: Male
Case Number: TN25-166426
Diagnosis: Non-small cell carcinoma

Specimen Information

Primary Tumor Site: Lung, NOS
Specimen Site: Thoracic lymph node
Specimen ID: C25-04482-A1
Specimen Collected: 15-Apr-2025
Test Report Date: 27-Apr-2025

Ordered By

Janeesh Sekkath-Veedu, MD
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(859) 257-4488

Results with Therapy Associations

BIOMARKER	METHOD	ANALYTE	RESULT	THERAPY ASSOCIATION		BIOMARKER LEVEL*
PD-L1 (22c3)	**IHC**	**Protein**	**Positive, TPS: 98%**	BENEFIT	**cemiplimab, pembrolizumab**	**Level 1**
PD-L1 (28-8)	**IHC**	**Protein**	**Positive | 2+, 98%**	BENEFIT	**nivolumab/ipilimumab combination**	**Level 1**
PD-L1 (SP142)	**IHC**	**Protein**	**Positive, TC: 2+, 90%**	BENEFIT	**atezolizumab (metastatic)**	**Level 1**
PD-L1 (SP263)	**IHC**	**Protein**	**Positive, TC: 2+, 95%**	BENEFIT	**atezolizumab (adjuvant), cemiplimab**	**Level 1**
KRAS	**Seq**	**DNA-Tumor**	**Pathogenic Variant Exon 2 | p.G12C**	BENEFIT	**adagrasib, sotorasib**	**Level 2**
MET	IHC	Protein	Positive | 3+, 100%	BENEFIT	telisotuzumab vedotin	Level 3
	Seq	RNA-Tumor	Variant Transcript Not Detected	LACK OF BENEFIT	capmatinib, tepotinib	Level 2
					crizotinib	Level 2
ALK	IHC	Protein	Negative | 0	LACK OF BENEFIT	alectinib, ceritinib, crizotinib, lorlatinib	Level 1
					brigatinib	Level 2
	Seq	RNA-Tumor	Fusion Not Detected		alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	Level 2
BRAF	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	Level 2
EGFR	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	afatinib, dacomitinib, erlotinib, gefitinib, osimertinib	Level 2
RET	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	pralsetinib, selpercatinib	Level 2
ROS1	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	ceritinib, crizotinib, entrectinib, lorlatinib, repotrectinib	Level 2

* Biomarker reporting classification - Level 1 – MI Cancer Seek™ or other companion diagnostic (CDx) performed as part of professional services; Level 2 – Strong evidence of clinical significance or is endorsed by standard clinical guidelines; Level 3 – Potential clinical significance. Bolded benefit therapies, if present, highlight the most clinically significant findings.

The selection of any, all, or none of the matched therapies resides solely with the discretion of the treating physician. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, the FDA prescribing information for any therapeutic, and in accordance with the applicable standard of care. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. All trademarks and registered trademarks are the property of their respective owners.

Important Note

This content is provided as part of the professional services. For MI Cancer Seek[TM] results, see CDx Associated Findings.

This patient has a potential NCI-ComboMATCH Trial-eligible result. Please see the clinical trials section on *Page 6*.

Immune Cell staining for PD-L1 SP142 cannot be reported in fine needle aspiration or body fluid specimens due to absence of tumor associated stroma.

This patient's sample is positive for expression of MET by immunohistochemistry (Ventana, SP44) and is associated with telisotuzumab vedotin, pending FDA approval (NCT04928846). Please see clinicaltrials.gov for additional trials which may require different thresholds for MET positivity (e.g. NCT05652868, NCT03175224, NCT03797391,NCT06083857, NCT05845671).

Cancer-Type Relevant Biomarkers

Biomarker	Method	Analyte	Result
TP53	**Seq**	**DNA-Tumor**	**Pathogenic Variant Exon 8 | p.R273C**
MSI	Seq	DNA-Tumor	Stable
NTRK1/2/3	Seq	RNA-Tumor	Fusion Not Detected
Tumor Mutational Burden	Seq	DNA-Tumor	Low, 4 mut/Mb
ALK	Seq	DNA-Tumor	Mutation Not Detected
BRAF	Seq	RNA-Tumor	Fusion Not Detected
ERBB2 (Her2/Neu)	CNA-Seq	DNA-Tumor	Amplification Not Detected
	IHC	Protein	Negative | Score 0
	Seq	DNA-Tumor	Mutation Not Detected
FGFR3	Seq	RNA-Tumor	Fusion Not Detected

Biomarker	Method	Analyte	Result
KEAP1	Seq	DNA-Tumor	Mutation Not Detected
MET	CNA-Seq	DNA-Tumor	Intermediate
	Seq	DNA-Tumor	Mutation Not Detected
MTAP	CNA-Seq	DNA-Tumor	Deletion Not Detected
NFE2L2	Seq	DNA-Tumor	Mutation Not Detected
NRG1	Seq	RNA-Tumor	Fusion Not Detected
PD-L1 (SP142)	IHC	Protein	IC: See Notes
RB1	Seq	DNA-Tumor	Mutation Not Detected
RET	Seq	DNA-Tumor	Mutation Not Detected
STK11	Seq	DNA-Tumor	Mutation Not Detected

Genomic Signatures

Biomarker	Method	Analyte	Result
Microsatellite Instability (MSI)	Seq	DNA-Tumor	Stable
Tumor Mutational Burden (TMB)	Seq	DNA-Tumor	Result: Low 4 Low 10 High
Genomic Loss of Heterozygosity (LOH)	Seq	DNA-Tumor	Equivocal - 11% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

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Genes Tested with Pathogenic or Likely Pathogenic Alterations

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
ARID2	Seq	DNA-Tumor	Pathogenic Variant	p.G1699*	18	c.5095G>T	42
ASXL1	Seq	DNA-Tumor	Pathogenic Variant	p.I617fs	13	c.1848 _1849 insT	6
KRAS	Seq	DNA-Tumor	Pathogenic Variant	p.G12C	2	c.34G>T	45
TP53	Seq	DNA-Tumor	Pathogenic Variant	p.R273C	8	c.817C>T	36

Unclassified alterations for DNA and RNA sequencing can be found in the MI Portal.
Formal nucleotide nomenclature and gene reference sequences can be found in the Appendix of this report.
Variants of Uncertain Significance can be found in the MI Portal.

Predicted Human Leukocyte Antigen (HLA) Genotype Results

The predicted HLA genotype results are summarized in the table below. Please note that HLA typing via tumor tissue sequencing is not a substitute for histocompatibility testing performed using peripheral blood. It is recommended to confirm the patient's HLA type with an appropriate assay.

Gene	Method	Analyte	Genotype
MHC CLASS I			
HLA-A	Seq	DNA-Tumor	A*01:01, A*11:01
HLA-B	Seq	DNA-Tumor	B*08:01, B*15:01
HLA-C	Seq	DNA-Tumor	C*03:04, C*07:01

HLA genotypes with only one allele are either homozygous or have loss-of-heterozygosity at that position.

Immunohistochemistry Results

Biomarker	Result	Biomarker	Result
ALK	Negative | 0	PD-L1 (28-8)	Positive | 2+, 98%
ERBB2 (Her2/Neu)	Negative | Score 0	PD-L1 (SP142)	IC: See Notes Positive, TC: 2+, 90%
MET	Positive | 3+, 100%	PD-L1 (SP263)	Positive, TC: 2+, 95%
PD-L1 (22c3)	Positive, TPS: 98%		

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Genes Tested with Indeterminate Results by Tumor DNA Sequencing

ATP6AP2	CUL3	EED	JAK2	MDH2	NOTCH2	PMS1	PRKD1	REST	RRAS2	SSBP1	TRIM28
B2M	CYSLTR2	EGLN1	KDM6A	MDM2	NPM1	POLD3	PTPRD	RIT1	RXRA	STAG2	XRCC2
CBL	DACH1	EIF1AX	KIF1B	MGA	PARP1	POLQ	RABL3	ROS1	SMARCA2	SUZ12	YES1
COL2A1	DGCR8	FYN	LYN	MRE11	PLCB4	PREX2	RASA1	RPA2	SOS1	TGFBR2	

Genes in this table were ruled indeterminate due to low coverage for some or all exons.

Genes Tested with Intermediate CNA Results by Tumor DNA Sequencing

CDK6	MET	RAC1									

The results in this report were curated to represent biomarkers most relevant for the submitted cancer type. These include results important for therapeutic decision-making, as well as notable alterations in other biomarkers known to be involved in oncogenesis. Additional results, including genes with normal findings, variants of uncertain significance, or unclassified alterations can be found in the MI Portal at **miportal.carismolecularintelligence.com.** If you do not have an MI Portal account, or need assistance accessing it, please contact Caris Customer Support at (888) 979-8669.

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Notes of Significance

SEE APPENDIX FOR DETAILS
CDx reports are generated through automated bioinformatics processing of a pre-defined list of genes and variants. Pathogenic or likely pathogenic variants identified outside the FDA-approved list will not be included in the CDx report. In addition, downgrading of variants approved by board-eligible/board-certified geneticists and pathologists may have occurred through manual variant annotation carried out by Caris' professional services. To ensure optimal coverage, multiple sequencing runs may be performed for a given case. This may account for some discrepancies observed between CDx reports and professional services.
Immune Cell staining for PD-L1 SP142 cannot be reported in fine needle aspiration or body fluid specimens due to absence of tumor associated stroma.
Clinical Trials Connector ™ opportunities based on biomarker expression: 596 Targeted Therapies. See page 6 for details.

Specimen Information

Specimen ID: C25-04482-A1

Specimen Collected: 04/15/2025

Specimen Received: 04/18/2025

Testing Initiated: 04/22/2025

Test Ordered*: MI Profile™ (MI Cancer Seek™ + IHCs and Other Tests by Tumor Type)
* If the submitted specimen is inadequate, only a subset of the ordered testing may be reported.

Gross Description: 1 (A) Paraffin Block - Client ID(C25-04482-A1) from UK HealthCare - Albert B Chandler Hospital, Lexington, KY, with the corresponding pathology report labeled "C25-04482".

Dissection Information: A laboratory technician harvested targeted tissues for extraction from the marked areas using a dissection microscope.

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Clinical Trials Connector™

The Clinical Trials Connector lists agents that are matched to available clinical trials according to biomarker status. In some instances, older-generation agents may still be relevant in the context of new combination strategies and, therefore, will still appear on this report.

Therapeutic agents listed below may or may not be currently FDA approved for the tumor type tested.

Please see https://clinicaltrials.gov/ for more information.

NCI-COMBOMATCH BIOMARKER SUMMARY				
Description	**Biomarker**	**Method**	**Analyte**	**Investigational Agent(s)**
KRAS G12C mutation / sotorasib + panitumumab	KRAS	Seq	DNA-Tumor	sotorasib + panitumumab

The patient may be eligible to enroll in a treatment study under NCI-ComboMATCH. Regarding copy number analysis (CNA) criteria: NCI-ComboMATCH gene amplification thresholds are higher than the Caris reporting thresholds. As a result, only genes with amplification levels above the NCI-ComboMATCH threshold are shown in the table above. NCI-ComboMATCH gene deletion thresholds include both homozygous and heterozygous deletions, whereas Caris only reports homozygous deletions. As a result, only genes with homozygous deletions are shown in the table above. Please note, matching to a ComboMATCH arm in the table above does not guarantee enrollment to a treatment study, as some treatment trials have additional inclusion criteria. Further evaluation by the ComboMATCH protocol team will be necessary. Please contact NCI for confirmation.

TARGETED THERAPIES (596)				
Drug Class	**Biomarker**	**Method**	**Analyte**	**Investigational Agent(s)**
Cancer vaccines (1)	KRAS	NGS	DNA-Tumor	OSE2101
cMET-targeted therapy (39)	MET	IHC	Protein	brigatinib, cabozantinib, capmatinib, crizotinib, savolitinib, telisotuzumab vedotin, tepotinib
Dual MEK/RAF inhibitor (3)	KRAS	NGS	DNA-Tumor	avutometinib
Heat shock pathway inhibitors (1)	ARID2	NGS	DNA-Tumor	NXP800
Immunomodulatory agents (449)	PD-L1	IHC	Protein	atezolizumab, avelumab, balstilimab, cemiplimab, cetrelimab, dostarlimab, durvalumab, envafolimab, ipilimumab, nivolumab, pembrolizumab, retifanlimab, spartalizumab, tislelizumab, toripalimab, tremelimumab, XmAb20717, zimberelimab
	PD-L1 (SP142)	IHC	Protein	
KRAS G12C inhibitors (35)	KRAS	NGS	DNA-Tumor	adagrasib, BI 1701963, D-1553, GDC-6036, JAB-21822, LY3537982, sotorasib, YL-15293
MEK inhibitors (28)	KRAS	NGS	DNA-Tumor	binimetinib, cobimetinib, mirdametinib, trametinib
PARP inhibitors (31)	ARID2	NGS	DNA-Tumor	BGB-290, niraparib, olaparib, pamiparib, Saruparib (AZD5305), senaparib, talazoparib
SHP2 inhibitors (7)	KRAS	NGS	DNA-Tumor	HBI-2376, JAB-3312, PF-07284892, RMC-4630, TNO155
SWI/SNF complex inhibitors (2)	ARID2	NGS	DNA-Tumor	PRT3789

( ) = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

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Disclaimer

Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, prescribing information for any therapeutic, and in accordance with the applicable standard of care. Drug associations provided in this report do not guarantee that any particular agent will be effective for the treatment of any patient or for any particular condition. Caris Life Sciences® expressly disclaims and makes no representation or warranty whatsoever relating, directly or indirectly, to the performance of services, including any information provided and/or conclusions drawn from therapies that are included or omitted from this report. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. The selection of therapy, if any, resides solely in the discretion of the treating physician and the tests should not be considered a companion diagnostic.

Caris MPI, Inc. d/b/a Caris Life Sciences is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing, including all Caris molecular profiling assays. Individual assays that are available through Caris molecular profiling include both Laboratory Developed Tests (LDT) and U.S. Food and Drug Administration (FDA) approved or cleared tests. In addition, certain tests have been CE-marked as a general IVD under the In Vitro Diagnostic Directive (IVDD) 98/79/EC. Offered LDTs were developed and their performance characteristics determined by Caris. Certain tests have not been cleared or approved by the FDA. Caris LDTs are used for clinical purposes. They are not investigational or for research. Caris' CLIA certification number is located at the bottom of each page of this report.

The information presented in the Clinical Trials Connector™ section of this report, if applicable, is compiled from sources believed to be reliable and current. However, the accuracy and completeness of the information provided herein cannot be guaranteed. The clinical trials information present in the biomarker description was compiled from www.clinicaltrials.gov. The contents are to be used only as a guide, and health care providers should employ their best comprehensive judgment in interpreting this information for a particular patient. Specific eligibility criteria for each clinical trial should be reviewed as additional inclusion criteria may apply.

All materials, documents, data, data software, information and/or inventions supplied to customers by or on behalf of Caris or created by either party relating to the services shall be and remain the sole and exclusive property of Caris. Customer shall not use or disclose the information provided by Caris through the services or related reports except in connection with the treatment of the patient for whom the services were ordered and shall not use such property for, or disseminate such property to, any third parties without expressed written consent from Caris. Customer shall deliver all such property to Caris immediately upon demand or upon Caris ceasing to provide the services. The technical and professional component of all testing was performed at the laboratory location displayed in the footer unless otherwise notated in the report.

Caris molecular testing is subject to Caris' intellectual property. Patent www.CarisLifeSciences.com/ip.

Kurt B. Hodges

Electronic Signature
Kurt Hodges, MD, MBA
27-Apr-2025

(01)00860008613325(21)H5C33DSXF

MI Cancer Seek™

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CARIS | MI CANCER SEEK

Patient

Name: CARRIGAN, GARY L
Date of Birth: 06/19/1950
Sex: Male
Case Number: TN25-166426
CDx Report Generated: 04/26/2025

Specimen Information

Diagnosis and Tumor Type:
Non-small cell carcinoma, Lung, NOS
Specimen Site: Thoracic lymph node
Specimen ID: C25-04482-A1
Specimen Type: Formalin-fixed paraffin-embedded
Specimen Collected: 04/15/2025

Ordered By

Janeesh Sekkath-Veedu, MD
UK HealthCare - Albert B Chandler Hospital - Markey Cancer Center

CDx Associated Findings

Genomic Findings Detected	FDA-approved Therapeutic Options
Findings did not yield any biomarker results with Companion Diagnostic (CDx) Claims.	
EGFR L858R	Not Detected
EGFR exon 19 deletions	Not Detected
Microsatellite Instability (MSI)	Not MSI-H

Tumor Profiling Results

MI Cancer Seek is FDA-approved to provide tumor mutation profiling results for previously diagnosed oncology patients with solid tumors.

Other Alterations and Biomarkers Identified

Results reported in this section are not prescriptive or conclusive for labeled use of any specific therapeutic product.
Confirmation of tumor mutation status using an FDA-approved CDx test is needed for therapeutic use.

Tumor Mutational Burden (TMB)	4 mut/Mb
ARID2	G1699*
ASXL1	I617fs
KRAS	G12C
TP53	R273C

MI Cancer Seek™

Intended Use:

MI Cancer Seek is a next-generation sequencing (NGS) based in vitro diagnostic (IVD) device using total nucleic acid (TNA) isolated from formalin-fixed paraffin-embedded (FFPE) tumor tissue specimens for the detection of single nucleotide variants (SNVs) and insertions and deletions (indels) in 228 genes, microsatellite instability (MSI), tumor mutational burden (TMB) in patients with previously diagnosed solid tumors, and copy number amplification (CNA) in one gene in patients with breast cancer.

MI Cancer Seek is intended as a companion diagnostic to identify patients who may benefit from treatment with the targeted therapies listed in Table 1 below, in accordance with the approved therapeutic product labeling.

Additionally, MI Cancer Seek is intended to provide tumor mutational profiling to be used by qualified healthcare professionals in accordance with professional oncology guidelines for cancer patients with previously diagnosed solid malignant neoplasms. Genomic findings other than those listed in Table 1 are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Table 1. MI Cancer Seek Companion Diagnostic Indications

INDICATION	BIOMARKER	THERAPY
Breast Cancer	*PIK3CA* (C420R, E542K, E545A, E545D [1635G>T only], E545G, E545K, Q546E, Q546R, H1047L, H1047R, H1047Y)	PIQRAY® (alpelisib)
Colorectal Cancer (CRC)	*KRAS* wild-type (absence of mutations in exons 2, 3, and 4) and *NRAS* wild-type (absence of mutations in exons 2, 3, and 4)	VECTIBIX® (panitumumab)
	BRAF V600E	BRAFTOVI® (encorafenib) in combination with ERBITUX® (cetuximab)
Melanoma	*BRAF* V600E	BRAF Inhibitors approved by FDA*
	BRAF V600E or *BRAF* V600K	MEKINIST® (trametinib) or BRAF/MEK Inhibitor Combinations approved by FDA*
Non-small cell lung cancer (NSCLC)	*EGFR* exon 19 deletions and exon 21 L858R alterations	EGFR Tyrosine Kinase Inhibitors approved by FDA*
Solid Tumors	MSI-H	KEYTRUDA® (pembrolizumab), JEMPERLI (dostarlimab-gxly)
Endometrial Carcinoma	Not MSI-H	KEYTRUDA® (pembrolizumab) in combination with LENVIMA® (lenvatinib)
*For the most current information about the therapeutic products in this group, go to: https://www.fda.gov/medical-devices/in-vitro-diagnostics/list-cleared-or-approved-companion-diagnostic-devices-in-vitro-and-imaging-tools#Group_Labeling		

MI Cancer Seek™ is a single-site assay performed at Caris Life Sciences, Phoenix, AZ.

Contraindications:

There are no known contraindications.

Warnings and Precautions:

Biopsy may pose a risk to the patient when archival tissue is not available for use with the assay. The patient's physician should determine whether the patient is a candidate for biopsy.

Limitations:

- For *in vitro* diagnostic use.
- CAUTION: Federal law restricts this device to sale by or on the order of a physician.
- The acceptable preparation method for MI Cancer Seek CDx specimens is FFPE. Other preparations have not been evaluated.
- The test is designed to report out somatic variants and is not intended to report germline variants.
- MI Cancer Seek requires a minimum tumor percentage of 20% for detection of alterations, with tumor content enrichment recommended for specimens with tumor percentage lower than 20%.
- Genomic findings other than those listed in the Companion Diagnostic Indications table are not prescriptive or conclusive for labeled use of any specific therapeutic product. Confirmation of tumor mutation status using an FDA-approved CDx test is needed for therapeutic use.
- A negative result does not rule out the presence of a mutation below the limits of detection of the assay.
- MI Cancer Seek is only approved for use with Caris Life Sciences pre-qualified Illumina NovaSeq 6000 instruments.
- The test is intended to be performed on specific serial number-controlled instruments by Caris Life Sciences.
- MI Cancer Seek does not report TMB for values lower than 3 mut/Mb as the accuracy of TMB values below 3 mut/Mb are unreliable.
- Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care in a given community.
- Based on the low positive percent agreement (PPA) in the accuracy study for ERBB2 copy number amplifications (CNAs) in breast cancer patients, this alteration may not be detected. Additional clinical investigation to confirm the presence of ERBB2 CNAs in the breast cancer patient's tumor with another FDA approved or cleared test is strongly recommended.
- Patients with breast cancer whose specimens have intermediate ERBB2 CNA status should be tested with another FDA approved or cleared test to ascertain ERRB2 CNA status in their tumor.
- Test may have reduced sensitivity and may yield false negative results in samples where necrotic tissue is >15%, melanin is >5%, fat cells are >10%.

Test Principle:

MI Cancer Seek is a single-site assay performed at Caris Life Sciences located at 4610 South 44th Place, Phoenix, AZ 85040. The test includes reagents, software, and procedures for testing of total nucleic acid (TNA) from formalin fixed paraffin embedded (FFPE) tumor tissue. The test uses a custom bait panel to measure all coding regions of the exome to detect and report SNVs and indels within 228 genes outlined in Table 2 across solid tumors and amplifications in ERBB2 in patients with breast cancer only. The test also detects MSI (determined from 3,210 genes) and whole exome based TMB in patients with solid tumors.

Table 2. MI Cancer Seek Reportable Gene List for SNVs and indels

ABL1	*BCL9*	*CDKN2A*	*ESR1*	*FLT1*	*JAK3*	*MET*	*NRAS*	*PPP2R2A*	*RUNX1*	*STK11*
ACVR1	*BCOR*	*CHEK1*	*EZH2*	*FLT3*	*KDM5C*	*MITF*	*NSD1*	*PRDM1*	*SDHA*	*SUFU*
AIP	*BLM*	*CHEK2*	*FANCA*	*FOXA1*	*KDM6A*	*MLH1*	*NSD2*	*PRKACA*	*SDHAF2*	*TCF7L2*
AKT1	*BMPR1A*	*CIC*	*FANCB*	*FOXL2*	*KDR*	*MLH3*	*NTHL1*	*PRKAR1A*	*SDHB*	*TERT*
AKT2	*BRAF*	*CREBBP*	*FANCC*	*FUBP1*	*KEAP1*	*MPL*	*NTRK1*	*PRKDC*	*SDHC*	*TET2*
AKT3	*BRCA1*	*CSF1R*	*FANCD2*	*GATA3*	*KIT*	*MRE11*	*NTRK2*	*PTCH1*	*SDHD*	*TMEM127*
ALK	*BRCA2*	*CTCF*	*FANCE*	*GNA11*	*KLF4*	*MSH2*	*NTRK3*	*PTEN*	*SETD2*	*TNFAIP3*
AMER1	*BRIP1*	*CTNNA1*	*FANCF*	*GNA13*	*KMT2A*	*MSH3*	*PALB2*	*PTPN11*	*SF3B1*	*TNFRSF14*
APC	*BTK*	*CTNNB1*	*FANCG*	*GNAQ*	*KMT2C*	*MSH6*	*PBRM1*	*RAC1*	*SMAD2*	*TP53*
AR	*CALR*	*CXCR4*	*FANCI*	*GNAS*	*KMT2D*	*MTOR*	*PDGFRA*	*RAD50*	*SMAD4*	*TRAF7*
ARAF	*CARD11*	*CYLD*	*FANCL*	*H3F3A*	*KRAS*	*MUTYH*	*PDGFRB*	*RAD51B*	*SMARCA4*	*TSC1*
ARID1A	*CBFB*	*DDR2*	*FANCM*	*H3F3B*	*LZTR1*	*MYC*	*PIK3CA*	*RAD51C*	*SMARCB1*	*TSC2*
ARID2	*CCND1*	*DICER1*	*FAS*	*HIST1H3B*	*MAP2K1*	*MYCN*	*PIK3CB*	*RAD51D*	*SMARCE1*	*U2AF1*
ASXL1	*CCND2*	*DNMT3A*	*FAT1*	*HNF1A*	*MAP2K2*	*MYD88*	*PIK3R1*	*RAD54L*	*SMO*	*VHL*
ATM	*CCND3*	*EGFR*	*FBXW7*	*HOXB13*	*MAP2K4*	*NBN*	*PIK3R2*	*RAF1*	*SOCS1*	*WRN*
ATRX	*CD79B*	*EP300*	*FGFR1*	*HRAS*	*MAP3K1*	*NF1*	*PIM1*	*RASA1*	*SOS1*	*WT1*
AXIN2	*CDC73*	*EPHA2*	*FGFR2*	*IDH1*	*MAPK1*	*NF2*	*PMS2*	*RB1*	*SPEN*	*XPO1*
B2M	*CDH1*	*ERBB2*	*FGFR3*	*IDH2*	*MAX*	*NFE2L2*	*POLD1*	*RET*	*SPOP*	*XRCC1*
BAP1	*CDK12*	*ERBB3*	*FGFR4*	*IRF4*	*MED12*	*NFKBIA*	*POLE*	*RHOA*	*SRC*	
BARD1	*CDK4*	*ERBB4*	*FH*	*JAK1*	*MEF2B*	*NOTCH1*	*POT1*	*RNF43*	*STAG2*	
BCL2	*CDKN1B*	*ERCC2*	*FLCN*	*JAK2*	*MEN1*	*NPM1*	*PPP2R1A*	*ROS1*	*STAT3*	

<u>FDA Evidence Levels</u>:

Genomic findings other than those listed in the Intended Use are not prescriptive or conclusive for labeled use of any specific therapeutic product. Test results should be interpreted in the context of pathological evaluation of tumors, treatment history, clinical findings, and other laboratory data. The test report includes genomic findings reported in the following levels (Table 3).

Table 3. Classification Criteria for FDA Evidence Levels

LEVEL	CRITERIA
Level 1	Biomarker is FDA-approved as a companion diagnostic as part of MI Cancer Seek™
Level 2	Cancer alterations that are well-established with strong clinical evidence that the clinician must know according to professional consensus guidelines in the specific tumor type.
Level 3	Cancer alterations with potential clinical significance, e.g., biomarkers deemed useful for directing patients to a clinical trial or simply for informational purposes.
	i. Clinical data such as case reports, single or several case series, or Phase I/II clinical trial data that support the utility of specific biomarker alteration to direct a patient to clinical trials, or ii. Pre-clinical and/or *in vitro* studies provide structural analysis of the mutation, fusion, or isoform to confirm pathogenicity (tumor-promoting), sensitivity, or resistance through colony forming assays, growth inhibition or drug sensitivity assays, etc.

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Gene Expression

Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type
ADORA2A	92	FOLR1	44	NTRK3	76
ALK	85	GPC3	12	PDCD1	91
ASCL1	89	HGF	5	PDCD1LG2	76
ATM	30	HRAS	60	PIK3CA	66
AURKA	87	IGF1R	4	POU3F2	88
BRAF	52	KDM1A	44	PRAME	71
BRCA1	80	KDR	3	PTEN	34
BRCA2	90	KEAP1	0	RB1	6
BRD4	100	KRAS	38	RET	84
CCND1	47	LAG3	91	ROR1	80
CCND2	31	MAGEA4	84	ROR2	44
CCNE1	60	MDM2	28	ROS1	22
CD274	97	MET	98	SLFN11	20
CD276	2	MKI67	95	SRC	50
CDKN2A	82	MSLN	77	SSTR2	82
CEACAM5	21	MTAP	67	SSTR3	76
CLDN18	3	MTOR	64	SSTR5	85
CLDN6	85	MUC1	58	STK11	20
CTLA4	90	MUC16	72	TACSTD2	94
DLL3	77	MYC	96	TGFB1	20
EGFR	55	NECTIN4	20	TOP1	78
EPHA2	96	NEUROD1	90	TP53	68
ERBB2	29	NF1	5	TSC1	71
ERBB3	24	NRAS	59	TSC2	57
FGFR1	9	NRG1	82	VEGFA	62
FGFR2	29	NTRK1	90	XPO1	57
FGFR3	53	NTRK2	64	YAP1	34

Gene Expression of Selected Genes by Whole Transcriptome Sequencing (WTS) Methods:
Gene expression is derived from whole transcriptome sequencing. Relative expression of genes are calculated as normalized values using Transcripts per Million Molecules or TPM. TPM is presented as a percentile derived by comparison to a distribution of Caris' internal cohort of the tumor-type profiled. Selected genes reported in this section were chosen based on their tumor-type specific relevance for matching to clinical trials, or tumor type subclassification.

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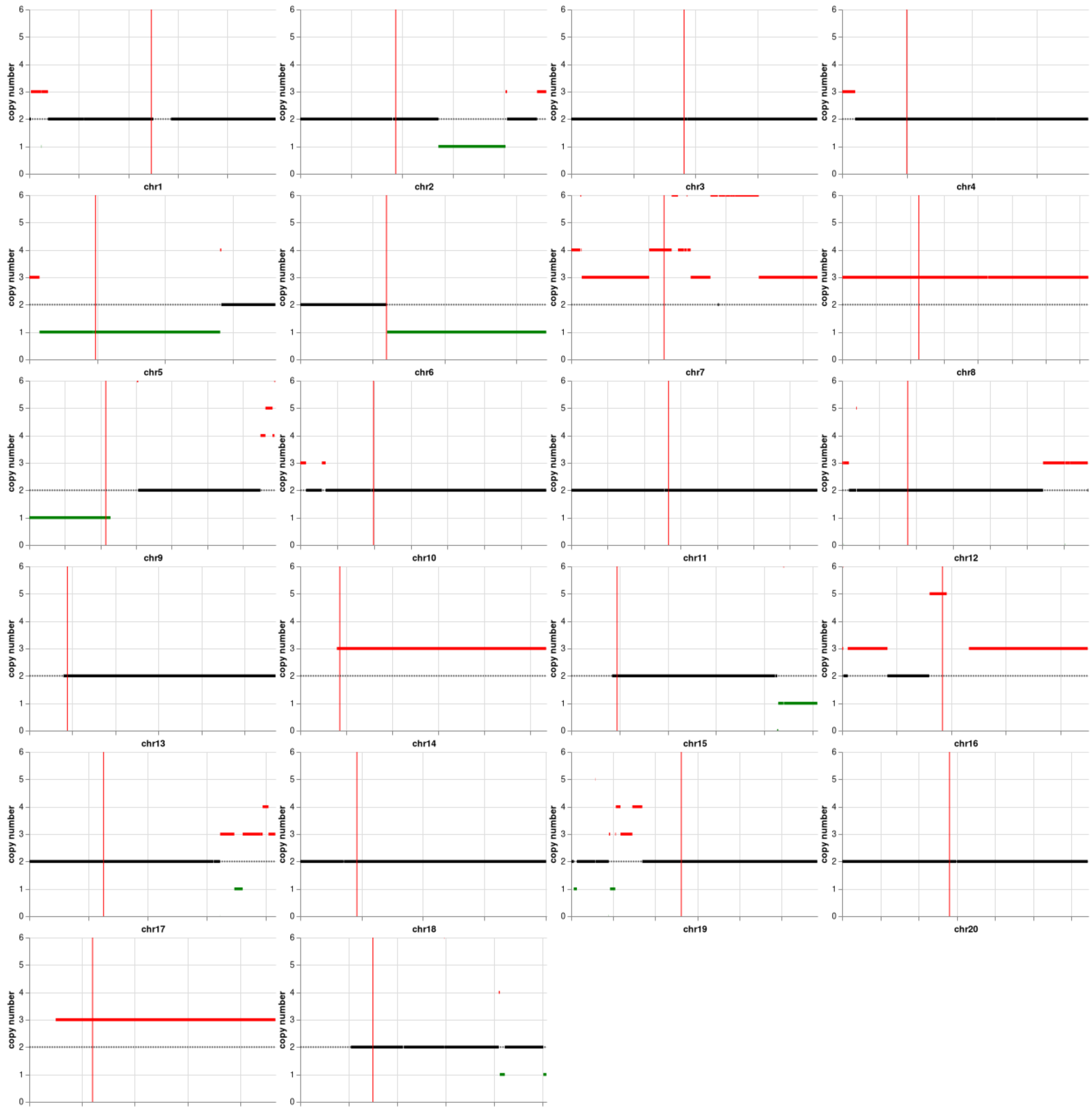
Karyotype



Karyotyping using Copy Number Analysis by Whole Exome Sequencing (WES) Methods:
Whole exome sequencing in combination with interrogation of single nucleotide polymorphisms (SNPs) tiled throughout the genome, allows for the identification and visualization of cytogenetic aberrations.

Somatic structural variants like whole or partial chromosome duplications or deletions, are important for cancer development and progression, and may identify clinically actionable alterations.

PATIENT: CARRIGAN, GARY (19-JUN-1950) **TN25-166426** **PHYSICIAN:** JANEESH SEKKATH-VEEDU, MD

Mutational Analysis by Next-Generation Sequencing (NGS)

TUMOR MUTATIONAL BURDEN	
Mutations / Megabase	Result
4	Low

TMB

Tumor Mutational Burden (TMB) is defined as the number of somatic non-synonymous mutations per million bases of sequenced DNA in a tumor sample. Tumors with high TMB may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. TMB analysis was performed based on next generation sequencing analysis of genomic DNA isolated from a tumor sample.

MICROSATELLITE INSTABILITY ANALYSIS	
Test	Result
MSI	Stable

MSI

Microsatellite instability (MSI) status is a measure of the number of somatic mutations within short, repeated sequences of DNA (microsatellites). MSI-High status can indicate that the tumor has a defect in mismatch repair (MMR) abrogating the ability to correct mistakes during DNA replication. Tumors with MSI-high status may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. Tumor-only microsatellite instability status by NGS (MSI-NGS) is measured by the direct analysis of known microsatellite regions sequenced in the CMI NGS panel.

GENOMIC LOSS OF HETEROZYGOSITY	
Test	Result
Genomic Loss of Heterozygosity (LOH)	Equivocal - 11% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

LOH

To calculate genomic loss-of-heterozygosity (LOH), the 22 autosomal chromosomes are split into 552 segments and the LOH of single nucleotide polymorphisms (SNPs) within each segment is calculated. Caris WES data consist of approximately 250k SNPs spread across the genome. SNP alleles with frequencies skewed towards 0 or 100% indicate LOH (heterozygous SNP alleles have a frequency of 50%). The final call of genomic LOH is based on the percentage of all 552 segments with observed LOH.

Additional Next-Generation Sequencing results continued on the next page. >

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Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
ARID2	DNA-Tumor	Pathogenic Variant	p.G1699*	18	c.5095G>T	42	NM_152641.3

Interpretation: A pathogenic mutation was detected in ARID2

ARID2 (AT-rich interactive domain 2) is a subunit of the PBAF chromatin-remodeling complex , which facilitates ligand-dependent transcriptional activation by nuclear receptors and repression of select genes by chromatin remodeling (alteration of DNA-nucleosome topology). It's required for the stability of the SWI/SNF chromatin remodeling complex SWI/SNF-B (PBAF). ARID2 is a significant tumor suppressor in many cancer types. Mutations are the most prevalent in hepatocellular cancer and melanoma, and are present in smaller but significant fraction in other tumors.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
ASXL1	DNA-Tumor	Pathogenic Variant	p.I617fs	13	c.1848 _1849 insT	6	NM_015338.5

Interpretation: A pathogenic frameshift mutation was detected in ASXL1

The protein is a member of the Polycomb group of proteins, which are necessary for the maintenance of stable repression of homeotic and other loci. The protein is thought to disrupt chromatin in localized areas, enhancing transcription of certain genes while repressing the transcription of other genes. The protein encoded by this gene functions as a ligand-dependent co-activator for retinoic acid receptor in cooperation with nuclear receptor coactivator 1. Mutations in this gene are associated with myelodysplastic syndromes and chronic myelomonocytic leukemia.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
KRAS	DNA-Tumor	Pathogenic Variant	p.G12C	2	c.34G>T	45	NM_004985.4

Interpretation: A pathogenic codon 12 mutation was detected in KRAS.

KRAS or V-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog encodes a signaling intermediate involved in many signaling cascades including the EGFR pathway. KRAS somatic mutations have been found in pancreatic (57%), colon (35%), lung (16%), biliary tract (28%), and endometrial (15%) cancers. Several germline mutations of KRAS (V14I, T58I, and D153V amino acid substitutions) are associated with Noonan syndrome.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TP53	DNA-Tumor	Pathogenic Variant	p.R273C	8	c.817C>T	36	NM_000546.5

Interpretation: A pathogenic mutation, p.R273C, was detected in TP53. Changes at codon 273 are frequent in cancers. p.R273C has also been reported as a germline mutation, causal for Li-Fraumeni syndrome (PMID: 21552135).

TP53, or p53, plays a central role in modulating response to cellular stress through transcriptional regulation of genes involved in cell-cycle arrest, DNA repair, apoptosis, and senescence. Inactivation of the p53 pathway is essential for the formation of the majority of human tumors. Mutation in p53 (TP53) remains one of the most commonly described genetic events in human neoplasia, estimated to occur in 30-50% of all cancers. Generally, presence of a disruptive p53 mutation is associated with a poor prognosis in all types of cancers, and diminished sensitivity to radiation and chemotherapy. Germline p53 mutations are associated with the Li-Fraumeni syndrome (LFS) which may lead to early-onset of several forms of cancer currently known to occur in the syndrome, including sarcomas of the bone and soft tissues, carcinomas of the breast and adrenal cortex (hereditary adrenocortical carcinoma), brain tumors and acute leukemias.

Additional Next-Generation Sequencing results continued on the next page. >

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Mutational Analysis by Next-Generation Sequencing (NGS)

GENES TESTED WITH INDETERMINATE* RESULTS BY TUMOR DNA SEQUENCING					
ATP6AP2	EED	MDH2	PMS1	REST	SSBP1
B2M	EGLN1	MDM2	POLD3	RIT1	STAG2
CBL	EIF1AX	MGA	POLQ	ROS1	SUZ12
COL2A1	FYN	MRE11	PREX2	RPA2	TGFBR2
CUL3	JAK2	NOTCH2	PRKD1	RRAS2	TRIM28
CYSLTR2	KDM6A	NPM1	PTPRD	RXRA	XRCC2
DACH1	KIF1B	PARP1	RABL3	SMARCA2	YES1
DGCR8	LYN	PLCB4	RASA1	SOS1	

*** Genes in this table were ruled indeterminate due to low coverage for some or all exons.**

For a complete list of genes tested, visit www.CarisMolecularIntelligence.com/profilemenu.

NGS Methods

Direct sequence analysis was performed on genomic DNA isolated from a micro-dissected tumor sample using Illumina NovaSeq 6000 sequencers. A hybrid pull-down panel of baits was used to enrich more than 700 clinically relevant genes along with > 20,000 other genes. Sequence data is analyzed using a customized bioinformatics pipeline to detect sequencing variants, copy number alterations (amplifications and deletions) indels and HLA genotypes. In addition, genomic signatures for tumor mutational burden (TMB), microsatellite instability (MSI), genomic loss-of-heterozygosity (LOH) or HRD-Genomic Scar Score (HRD-GSS), and homologous recombination deficiency (HRD) are reported when applicable. For a complete list of what is covered by the assay, and genes with partial coverage, please contact Caris Customer Support.

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Copy Number Alterations by Next-Generation Sequencing (NGS)

GENES TESTED WITH INTERMEDIATE CNA RESULTS					
CDK6	MET	RAC1			

CNA Methods

The copy number alteration (CNA) of each exon is determined by a calculation using the average sequencing depth of the sample along with the sequencing depth of each exon and comparing this calculated result to a pre-calibrated value. A complete list of genes for reporting copy number alterations, including amplifications and deletions, is available upon request.

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Gene Fusion and Transcript Variant Detection by RNA Sequencing

Whole Transcriptome Sequencing (WTS) Methods

Gene fusion and variant transcript detection were performed on RNA isolated from a tumor sample using next generation sequencing. The assay also detects fusions occurring at known and novel breakpoints within genes. The genes included in this report represent the subset of genes associated with cancer. The complete list of unclassified alterations is available by request.

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Protein Expression by Immunohistochemistry (IHC)

Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
ALK	0	100	Negative	Intensity ≥3+ and ≥1% of cells stained
MET	3 +	100	Positive	Intensity ≥3+ and ≥25% of cells stained

PD-L1 TUMOR CELL STAINING				
Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
PD-L1 (28-8)	2 +	98%	Positive	Intensity ≥1+ and ≥1% of cells stained
PD-L1 (SP142)	2 +	90%	Positive	Intensity ≥1+ and ≥50% of cells stained
PD-L1 (SP263)	2 +	95%	Positive	Intensity ≥1+ and ≥1% of cells stained

PD-L1 (28-8): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.
PD-L1 (SP142): TC scoring was based on the presence of discernible PD-L1 membrane staining of any intensity in ≥ 50% of viable tumor cells.
PD-L1 (SP263): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

PD-L1 TUMOR PROPORTION SCORE (TPS)			
Biomarker	Result	TPS	Threshold
PD-L1 (22c3)	Positive	98%	TPS ≥1 %

PD-L1 22c3: Scoring was based on the percentage of viable tumor cells showing partial or complete membrane staining. There are three categories of PD-L1 expression defined by the PD-L1 22c3 IHC pharmDx NSCLC interpretation guide: TPS < 1% (negative), TPS ≥ 1% and TPS ≥ 50%.

Her-2 IHC: Biopsy	
Final Score	Threshold for Positive Result
Negative | Score 0	Intensity =3+ and ≥5 tumor cells staining

PD-L1 IMMUNE CELL (IC) SCORE			
Biomarker	Result	IC	Threshold
PD-L1 (SP142)	See Below	N/A	≥10%

PD-L1 (SP142): IC scoring was based on discernible PD-L1 staining of any intensity in tumor-infiltrating immune cells covering ≥ 10% of tumor area occupied by tumor cells, associated intratumoral or contiguous peritumoral stroma.

Clones used: ALK (D5F3), ERBB2 (Her2/Neu) (4B5), MET (SP44) RxDx Assay, PD-L1 (22c3), PD-L1 (28-8), PD-L1 (SP142), PD-L1 (SP263).

Kurt B. Hodges

Electronic Signature
Kurt Hodges, MD, MBA
04/26/2025

Additional IHC results continued on the next page. >

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Protein Expression by Immunohistochemistry (IHC)

IHC Methods

The Laboratory Developed Tests (LDT) immunohistochemistry (IHC) assays were developed and their performance characteristics determined by Caris Life Sciences. These tests have not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not currently necessary. Interpretations of all immunohistochemistry (IHC) assays were performed manually or with the assistance of an AI-based image analysis tool by a board certified pathologist using a microscope and/or digital whole slide image(s).

The following IHC assays were performed using FDA-approved companion diagnostic or FDA-cleared tests consistent with the manufacturer's instructions: ALK (VENTANA ALK (D5F3) CDx Assay, Ventana), ER (CONFIRM anti-Estrogen Receptor (ER) (SP1), Ventana), FOLR1 (VENTANA FOLR1-2.1 RxDx, Ventana), CLDN18 (VENTANA, 43-14A RxDx Assay, Gastric/GEJ), PR (CONFIRM anti-Progesterone Receptor (PR) (1E2), Ventana), HER2/neu (PATHWAY anti-HER-2/neu (4B5), Ventana), Ki-67 (MIB-1 pharmaDx, Dako), MAGE-A4 1F9 (pharmDx, Dako), PD-L1 22c3 (pharmDx, Dako), PD-L1 SP142 (VENTANA, non-small cell lung cancer), PD-L1 28-8 (pharmDx, Dako, Gastric/GEJ, non-small cell lung cancer), PD-L1 SP263 (Ventana, non-small cell lung cancer), and Mismatch Repair (MMR) proteins (MLH1, MSH2, MSH6, and PMS2; VENTANA MMR RxDx Panel, Ventana).

HER2 results and interpretation follow the ASCO/CAP scoring criteria. Bartley, A.N., J.A. Ajani, et al. (2016). "HER2 testing and clinical decision making in gastroesophageal adenocarcinoma: guideline from the College of American Pathologists, American Society for Clinical Pathology, and the American Society of Clinical Oncology". J Clin Oncol. 35(4):446-464.

Comments on IHC Analysis

Immune Cell staining for PD-L1 SP142 cannot be reported in fine needle aspiration or body fluid specimens due to absence of tumor associated stroma.

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References

#	Drug	Biomarker	Reference
1	alectinib	ALK	Camidge, D.R., A.T. Shaw, et al. (2019). "Updated Efficacy and Safety Data and Impact of the EML4-ALK Fusion Variant on the Efficacy of Alectinib in Untreated ALK-Positive Advanced Non–Small Cell Lung Cancer in the Global Phase III ALEX Study." J Thorac Oncol 14(7): 1233-1243. View Citation Online
2	alectinib	ALK	Gadgeel, S., D.R. Camidge, et al. (2018). "Alectinib versus crizotinib in treatment-naive anaplastic lymphoma kinase-positive (ALK+) non-small-cell lung cancer: CNS efficacy results from the ALEX study." Ann Oncol 29 (11): 2214-2222. View Citation Online
3	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	Lindeman, N.I., Y. Yatabe, et al. (2018). "Updated Molecular Testing Guideline for the Selection of Lung Cancer Patients for Treatment With Targeted Tyrosine Kinase Inhibitors Guideline From the College of American Pathologists, the International Association for the Study of Lung Cancer, and the Association for Molecular Pathology." J Thorac Oncol 13(3): 323-358. View Citation Online
4	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology. Non-Small Cell Lung Cancer Version 1.2020
5	brigatinib	ALK	Camidge, D.R., S. Popat, et al. (2018). "Brigatinib versus Crizotinib in ALK-Positive Non-Small-Cell Lung Cancer." N Engl J Med 379(21): 2027-2039. View Citation Online
6	brigatinib	ALK	Lin, J.L., G.J. Riely, et al. (2018). "Brigatinib in Patients with Alectinib-refractory ALK-positive NSCLC." J Thorac Onc 13(10): 1530-1538. View Citation Online
7	brigatinib	ALK	Reckamp, K., J. Lee, et al. (2019). "Comparative efficacy of brigatinib versus ceritinib and alectinib in patients with crizotinib-refractory anaplastic lymphoma kinase-positive non-small cell lung cancer." Curr Med Res Opin 35(4):569-576. View Citation Online
8	brigatinib, crizotinib	ALK	Thorne-Nuzzo, T., P. Towne, et al. (2017). "A Sensitive ALK Immunohistochemistry Companion Diagnostic Test Identifies Patients Eligible for Treatment with Crizotinib." J Thorac Oncol 12(5): 804-813 View Citation Online
9	ceritinib, lorlatinib	ALK	Shaw, A.T., E. Felip, et al. (2017). "Ceritinib versus chemotherapy in patients with ALK-rearranged non-small-cell lung cancer previously given chemotherapy and crizotinib (ASCEND-5): a randomised, controlled, open-label, phase 3 trial." Lancet Oncol 18 (7):874-886. View Citation Online
10	ceritinib, lorlatinib	ALK	Solomon, B. J., A. T. Shaw, et al. (2018). "Lorlatinib in patients with ALK-positive non-small cell lung cancer: results from a global phase 2 study." Lancet Oncol 19:1654-1667. View Citation Online
11	ceritinib, lorlatinib	ALK	Soria, J.C., de Castro G., et al. (2017). "First-line ceritinib versus platinum-based chemotherapy in advanced ALK-rearranged non-small-cell lung cancer (ASCEND-4): a randomised, open-label, phase 3 study." Lancet 389: 917-929. View Citation Online
12	crizotinib	ALK	Solomon, B. J., T. S. Mok, et al. (2018). "Final overall survival analysis from a study comparing first-line crizotinib versus chemotherapy in ALK-mutation positive Non-small-cell-lung cancer." J Clin Oncol 36: 2251-2258. View Citation Online
13	crizotinib	ALK	van der Wekken, A. J., H.J.M Groen, et al. (2017). "Dichotomous ALK IHC is a better predictor for ALK inhibition outcome than traditional ALK FISH in advanced Non-small cell lung cancer." Clin Cancer Res 23(15): 4251-4258. View Citation Online
14	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Hyman, D.H., J. Baselga, et al. (2015). "Vemurafenib in Multiple Nonmelanoma Cancers with BRAF V600 Mutations." NEJM 373(8):726-736. View Citation Online
15	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2016). "An open-label phase II trial of dabrafenib (D) in combination with trametinib (T) in patients (pts) with previously treated BRAF V600E–mutant advanced non-small cell lung cancer (NSCLC; BRF113928)." J Clin Oncol 34: 15_suppl, 107-107. View Citation Online
16	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2017). "Dabrafenib plus trametinib in patients with previously untreated BRAFV600E-mutant metastatic non-small-cell lung cancer: an open-label, phase 2 trial." Lancet Oncol 18(1):1307-1316. View Citation Online

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References

#	Drug	Biomarker	Reference
17	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Riely, G. J., B. E. Johnson, et al. (2023). "Phase II, Open-Label Study of Encorafenib Plus Binimetinib in Patients With BRAFV600-Mutant Metastatic Non–Small-Cell Lung Cancer." J Clin Oncol. 41 (21): 3700-3711 View Citation Online
18	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Subbiah, V., D.M. Hyman, et al. (2019). "Efficacy of Vemurafenib in Patients With Non–Small-Cell Lung Cancer With BRAF V600 Mutation: An Open-Label, Single-Arm Cohort of the Histology-Independent VE-BASKET Study." JCO Precis Oncol 2019: 3, 1-9. View Citation Online
19	afatinib, dacomitinib, osimertinib	EGFR	Sequist, L.V., M. Schuler, et al. (2013). "Phase III Study of Afatinib or Cisplatin Plus Pemetrexed in Patients with Metastatic Lung Adenocarcinoma With EGFR Mutations." J Clin Oncol ahead of print July 1, 2013, doi: 10.1200/JCO.2012.44.2806 View Citation Online
20	afatinib, dacomitinib, osimertinib	EGFR	Wu, Y-L, T.S. Mok, et al (2017). "Dacomitinib versus gefitinib as first-line treatment for patients with EGFR-mutation-positive non-small-cell lung cancer (ARCHER 1050): a randomized, open-label, phase III trial." Lancet Oncol. 2017;18(11):1454-1466. View Citation Online
21	afatinib, dacomitinib, osimertinib	EGFR	Yang, J.C., et al. (2013). "Symptom control and quality of life in LUX-Lung 3: a phase III study of afatinib or cisplatin/pemetrexed in patients with advanced lung adenocarcinoma with EGFR mutations." J Clin Oncol 31:3342-3350. View Citation Online
22	afatinib, dacomitinib, osimertinib	EGFR	Yang, J.C., et al. (2015). "Clinical activity of afatinib in patients with advanced non-small-cell lung cancer harbouring uncommon EGFR mutations: a combined post-hoc analysis of LUX-Lung 2, LUX-Lung 3, and LUX-Lung 6." Lancet Oncol; (7):830-838. View Citation Online
23	erlotinib, gefitinib	EGFR	Brugger, W., F. Cappuzzo, et. al. (2011). "Prospective molecular marker analyses of EGFR and KRAS from a randomized, placebo-controlled study of erlotinib maintenance therapy in advanced non-small-cell lung cancer." J. Clin. Oncol. 29:4113-4120. View Citation Online
24	erlotinib, gefitinib	EGFR	Fukuoka, M., T.S.K. Mok, et. al. (2011). "Biomarker analyses and final overall survival results from a phase III, randomized, open-label, first-line study of gefitinib versus carboplatin/paclitaxel in clinically selected patients with advanced non-small-cell lung cancer in Asia (IPASS). J. Clin. Oncol. DOI: 10.1200/JCO.2010.33.4235. View Citation Online
25	erlotinib, gefitinib	EGFR	Keedy, V.L., G. Gianconne, et. al. (2011). "American Society of Clinical Oncology Provisional Clinical Opinion: epidermal growth factor receptor (EGFR) mutation testing for patients with advanced non-small cell lung cancer considering first-line EGFR tyrosine kinase inhibitor therapy." J. Clin. Oncol. 29(15):2121-2127. View Citation Online
26	erlotinib, gefitinib	EGFR	Maemondo, M., T. Nukiwa, et. al. (2010). "Gefitinib or chemotherapy for non-small cell lung cancer with mutated EGFR." N. Engl. J. Med. 362:2380-8. View Citation Online
27	adagrasib, sotorasib	KRAS	Hong, D.S., B.T. Li, et al. (2020). "KRAS G12C Inhibition with Sotorasib in Advanced Solid Tumors." N Engl J Med. 383(13): 1207-1217. View Citation Online
28	adagrasib, sotorasib	KRAS	Janne, P. A., A. I. Spira, et al. (2022). "Adagrasib in Non-Small-Cell Lung Cancer Harboring a KRASG12C Mutation." N Engl J Med. 14;387(2): 120-131 View Citation Online
29	adagrasib, sotorasib	KRAS	Li, B.T, J. Wolf, et al. (2021). "CodeBreaK 100: Registrational Phase 2 Trial of Sotorasib in KRAS p.G12C Mutated Non-small Cell Lung Cancer". J Thorac Oncol. 16 (3): S61. View Citation Online
30	capmatinib, tepotinib	MET	Paik, P. K., X. Le, et al. (2020). "Tepotinib in patients with MET exon 14 (METex14) skipping advanced NSCLC: Updated efficacy results from VISION Cohort A." IASLC 2020 World Conference on Lung Cancer (Abstract: 1361) View Citation Online
31	capmatinib, tepotinib	MET	Paik, P.K., X. Le, et al. (2020). "Tepotinib in Non-Small Cell Lung Cancer with MET Exon 14 Skipping Mutations." N Engl N Med 383(10): 931-943. View Citation Online
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References

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